



Presented by:

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# Thesis Defense

Identifying Business States and Product States in  
Transaction Data

**Advisor**

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**Program**

Master of Science in Statistics and Applied Mathematics

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# Abstract

## 1. Context

- E-commerce Electronics Business
- January 5<sup>th</sup> 2020 - July 14<sup>th</sup> 2020
- 2162 products, 2 Million + Transactions

## 2. Transaction Data Components

- Distribution of transactions across products
- Variation in demand across products
- Behavior over time

## 3. State-based Framework

- Evaluate all three components simultaneously
  - **Business States** - Overview of operations
  - **Product States** - In-depth perspective of products





# Abstract

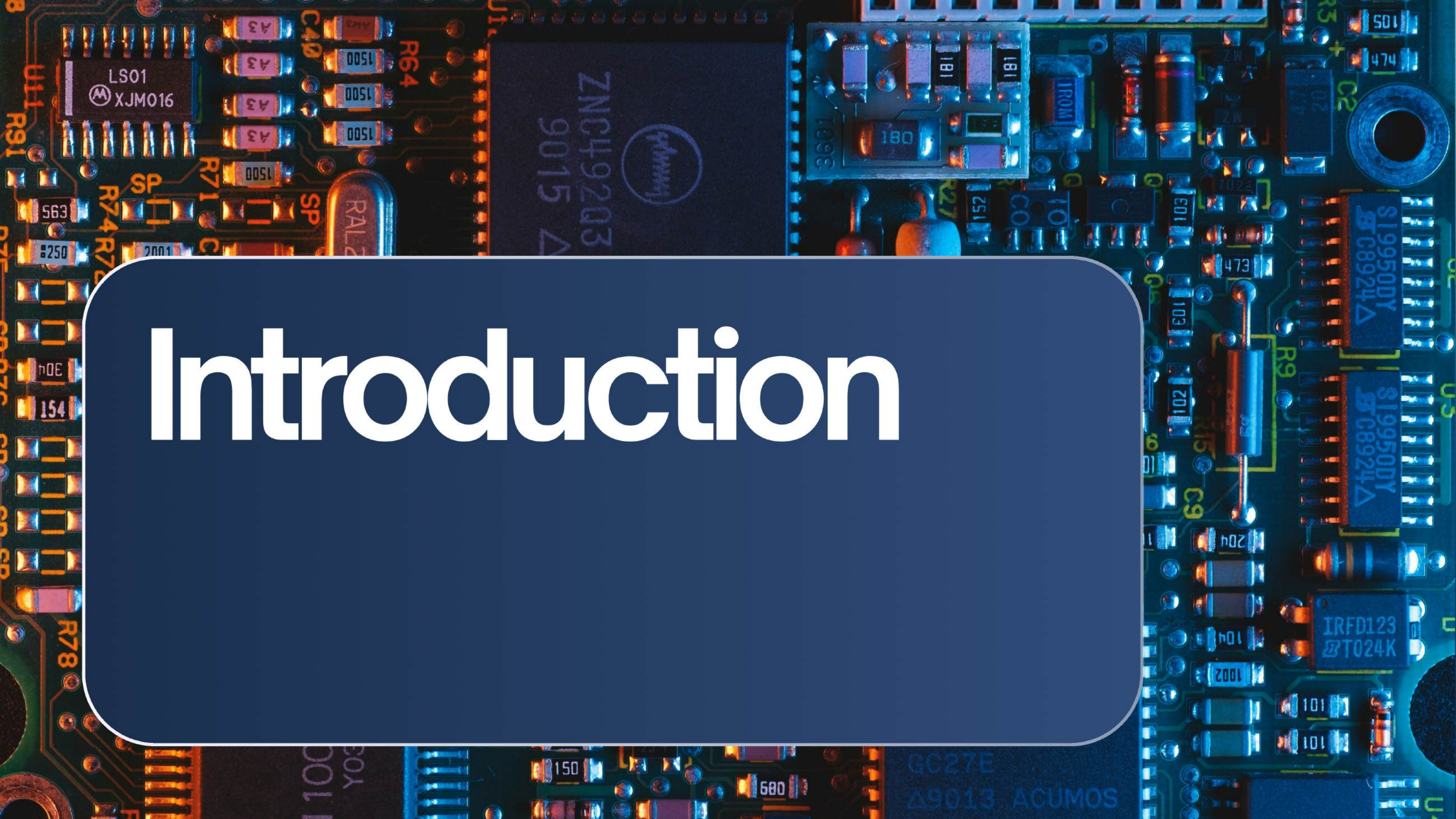
## 4. Methodology

- **Hierarchical Clustering**
  - Product Demand - Cluster volume by products
  - Product Preferences - Cluster by dates as product proportions
    - Product Trends - Subset product preferences by differences between products
- **Sequence Analysis**
  - Transitions of states
  - Interpretation of behavior

## 5. Outcomes

- Insights into operations and decision-making
- Inventory Strategies
- Marketing Initiatives





# Introduction



# Purpose

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## Proposed Research Statement

- How can daily transaction data be represented to capture product distribution, demand variation, and their temporal behavior over time?

## FRAMEWORK

- Identify product demand, product preferences, and product trend clusters
  - Model product trends to inform product behavior
- Characterize Business States and Product States
  - Analyze transitions and understand behavior over time
- Insights and recommendations
  - Operations, Customer Purchasing Patterns, Inventory Requirements, and Marketing Initiatives



# Terminology

- **Product preferences:**
  - Clusters of similar product distribution patterns by date regardless of volume
- **Product trends:**
  - Subgroups within product preferences that capture finer variations in product proportions across days
- **Product behavior:**
  - The temporal behavior of product trends, showing how purchasing patterns change or persist over time
- **Product demand:**
  - Clusters of similar number of product transactions by date regardless of product transaction shares

# Terminology

- **Business states:**

- Combined representation of product preferences and demand levels, describing overall operational conditions

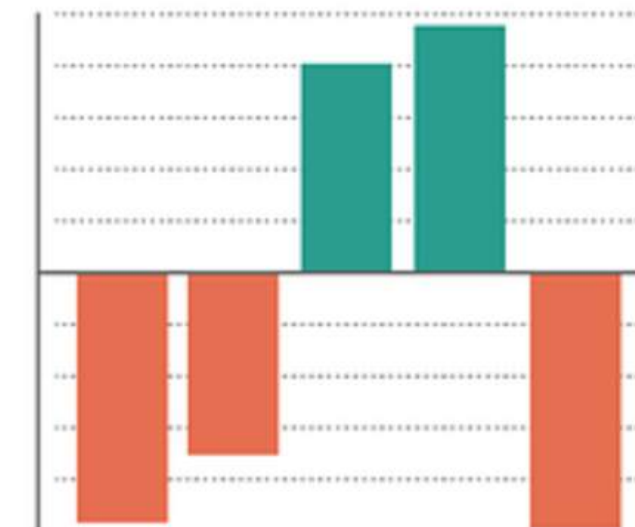
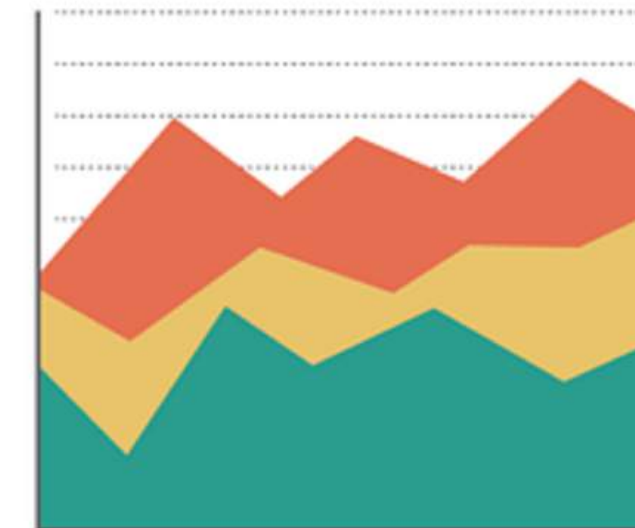
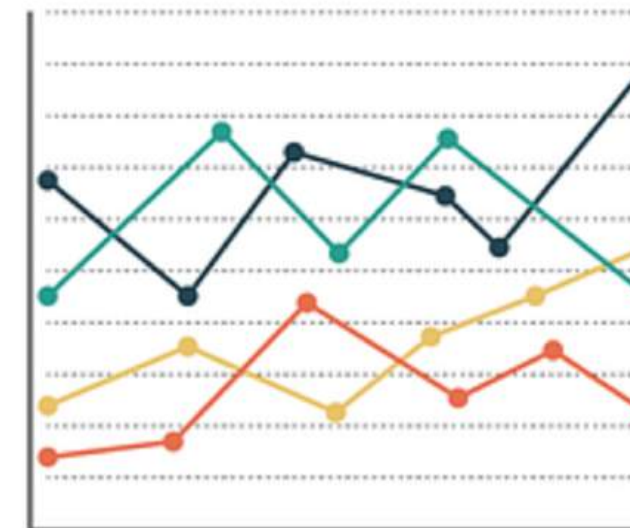
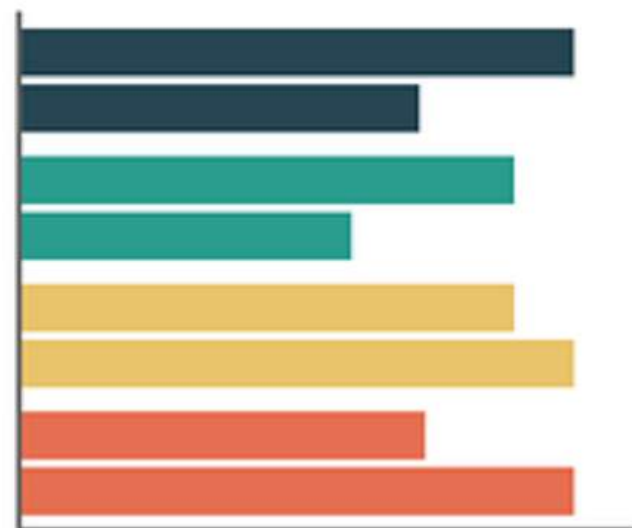
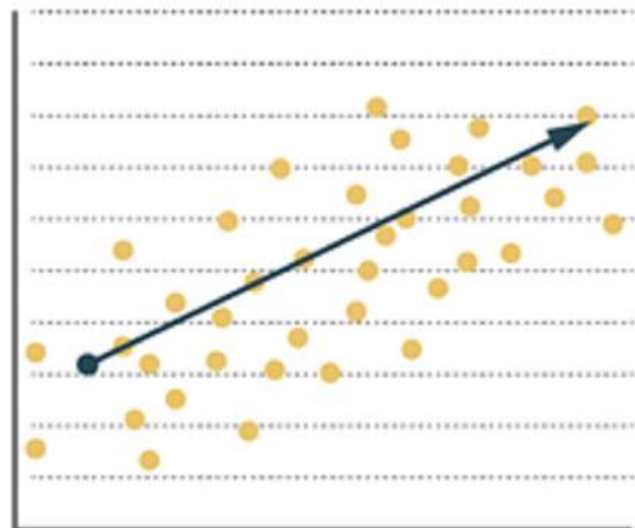
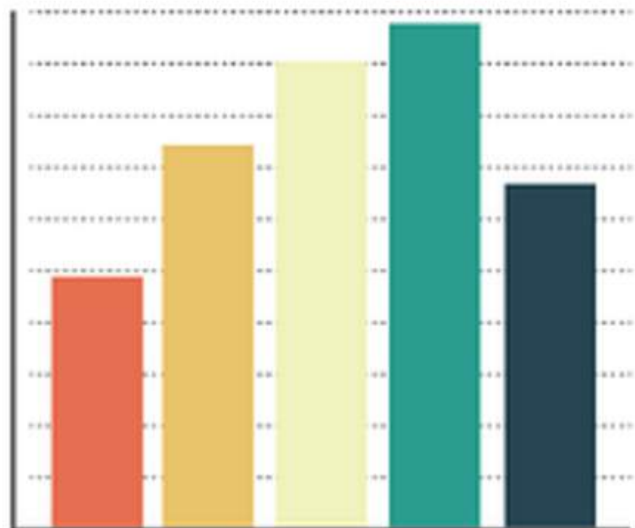
$$B_b = \{x_i \mid z_i^{(preferences)} = m, z_i^{(demand)} = d\}$$

- **Product states:**

- Combined representation of product trends and demand, revealing detailed purchasing patterns

$$P_r = \{x_i \mid z_i^{(trends)} = g, z_i^{(demand)} = h\}$$





# Context





# Data Overview

## Source

Kaggle

- Redacted from Open CDP

## Summary

2162 Products

2 million + transactions

Januray 5<sup>th</sup> 2020 - July 14<sup>th</sup> 2020

## Features

| Variable      | Description                            |
|---------------|----------------------------------------|
| Event Time    | Date and transaction time              |
| Order ID      | Unique identifier for each transaction |
| Product ID    | Unique identifier for each product     |
| Category Code | Description of the product             |
| Brand         | Name of the retailer                   |
| Price         | Product price                          |
| User ID       | Unique identifier for customers        |



# Processed Data

## Processing Steps

- Group transactions by Date and by Product
- Aggregate all products with mean < 10
  - 48 products + 1 “Other”
- Convert date: Day of week, Week in year

## Discrepancies

- One date 1/1/1970
- Only 1.04 million transactions used
- Incomplete Brand and Category data

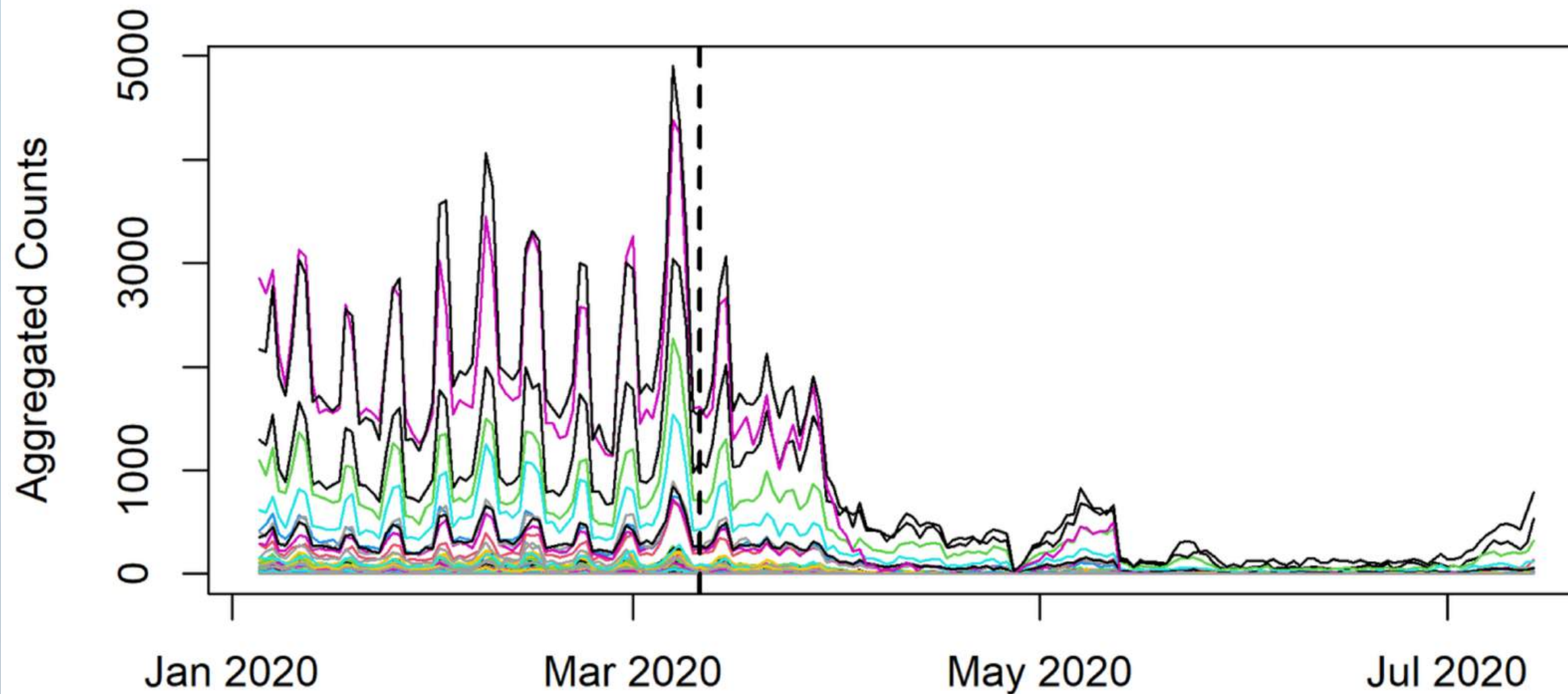
## Finalized Data

| Dates     | 1    | 2   | 3    | 4   |
|-----------|------|-----|------|-----|
| 1/1/1970  | 2291 | 470 | 1805 | 744 |
| 1/5/2020  | 1294 | 288 | 1097 | 418 |
| 1/6/2020  | 1246 | 252 | 954  | 476 |
| 1/7/2020  | 1540 | 317 | 1223 | 583 |
| 1/8/2020  | 1017 | 188 | 805  | 385 |
| 1/9/2020  | 882  | 187 | 783  | 336 |
| 1/10/2020 | 1255 | 254 | 1028 | 409 |
| 1/11/2020 | 1663 | 269 | 1367 | 571 |
| 1/12/2020 | 1501 | 291 | 1275 | 523 |
| 1/13/2020 | 861  | 205 | 769  | 355 |
| 1/14/2020 | 891  | 170 | 747  | 309 |
| 1/15/2020 | 817  | 180 | 724  | 316 |
| 1/16/2020 | 861  | 171 | 693  | 328 |



# Data Visualized

**Time Series of Aggregated Products by Date**



January 5th, 2020 - July 14th, 2020

**March 11<sup>th</sup>:**

WHO declares COVID-19 a  
Pandemic

- Labor Shortage
  - Unemployment rises to 13% (BLS, 2021)
- Supply Disruption
  - Canceled maritime shipments rises to 20% (USITC, 2021)



# Limitations

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## Limited Context

- Business unknown
- Product names, brand, categories unknown

## Aggregation Used

- Limits analysis to daily perspective
  - Customer behavior

## Inconsistent Data

- Missing transactions
  - Dates different than author stated
-



# Data Notation

## Observations (Continuous)

- Observations:  $n$ 
  - Days
- Variables:  $p$ 
  - Products

$$X = (x_{i,j}) \in \mathbb{N}^{n \times p},$$

## Proportions (Discrete)

Total Transactions by Day

$$m_{i.} = \sum_{j=1}^p x_{i,j}$$

→

Proportions Vector

$$\hat{\theta}_i = \left( \frac{x_{i1}}{m_{i.}}, \frac{x_{i2}}{m_{i.}}, \dots, \frac{x_{ip}}{m_{i.}} \right), \quad \hat{\theta}_i \in \Delta^{p-1}$$



# Data Notation

## Probability Simplex

- Set of all probability distributions
  - Composition of product proportions

$$\Delta^{p-1} = \left\{ \theta_i \in \mathbb{R}^p : \theta_{i,j} \geq 0, \sum_{j=1}^p \theta_{i,j} = 1 \right\}$$

## Clusters

Assignment

$$z_i \in \{1, \dots, K\}$$

Group Representation

$$\rightarrow C_k = \{i : z_i = k\}, \quad n_k = |C_k|$$



# Methodology

## CLUSTERING

- Group observations based on similarity
  - Choice in Dissimilarity Measure
- Validation of clusters

## MODELING

- Assume parametric distribution observations are drawn from
  - **Dirichlet** Distribution for proportions
  - Summary statistics for aggregated counts

## SEQUENCE ANALYSIS

- Understand behavior over time
  - Transitions between assignments
  - Reoccurring patterns



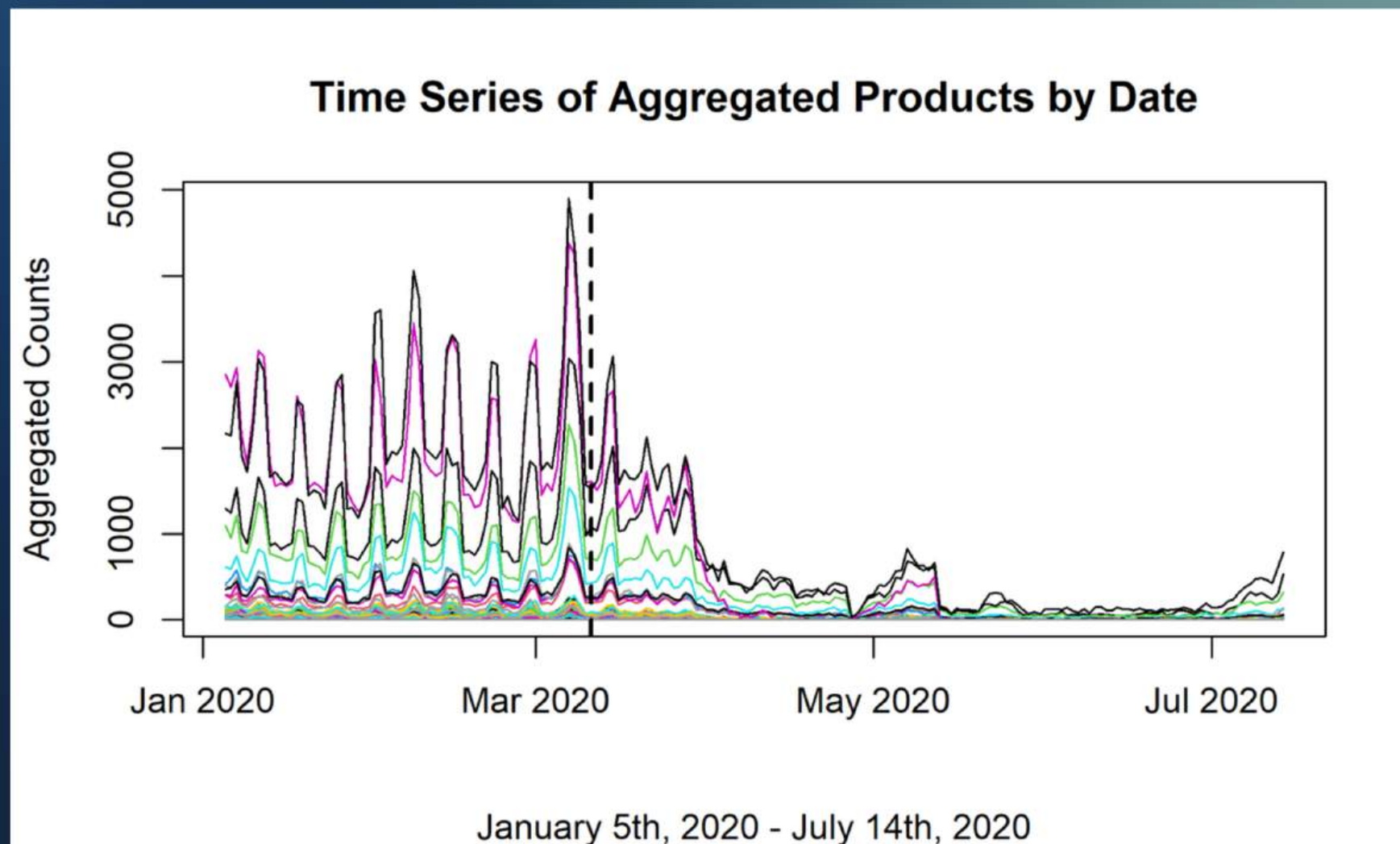
# Analysis



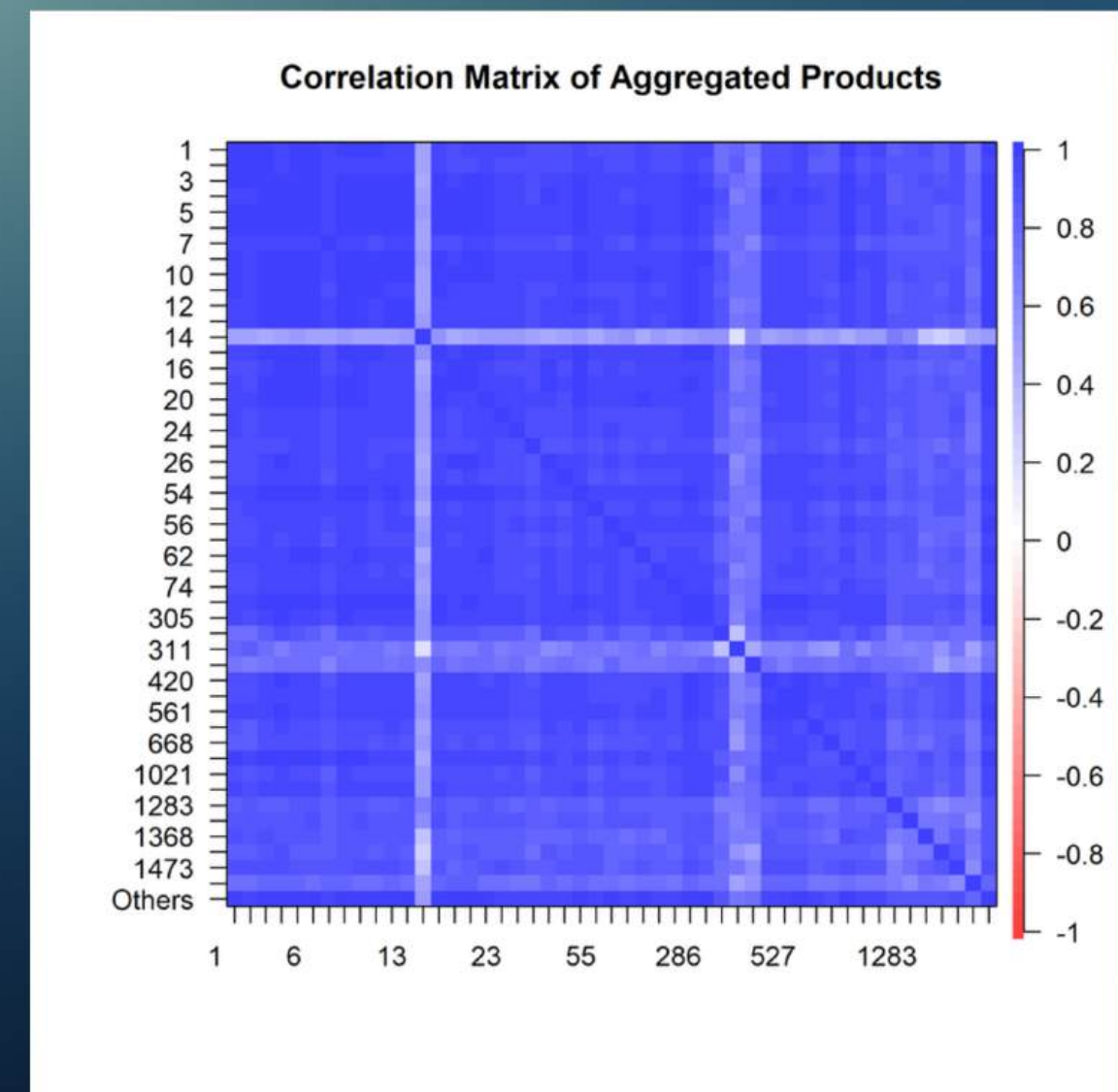
# Exploratory Analysis

**Purpose:** Understand what assumptions can be made about data

## Timeseries of Products



## Correlation Between Products





# Hierarchical Clustering

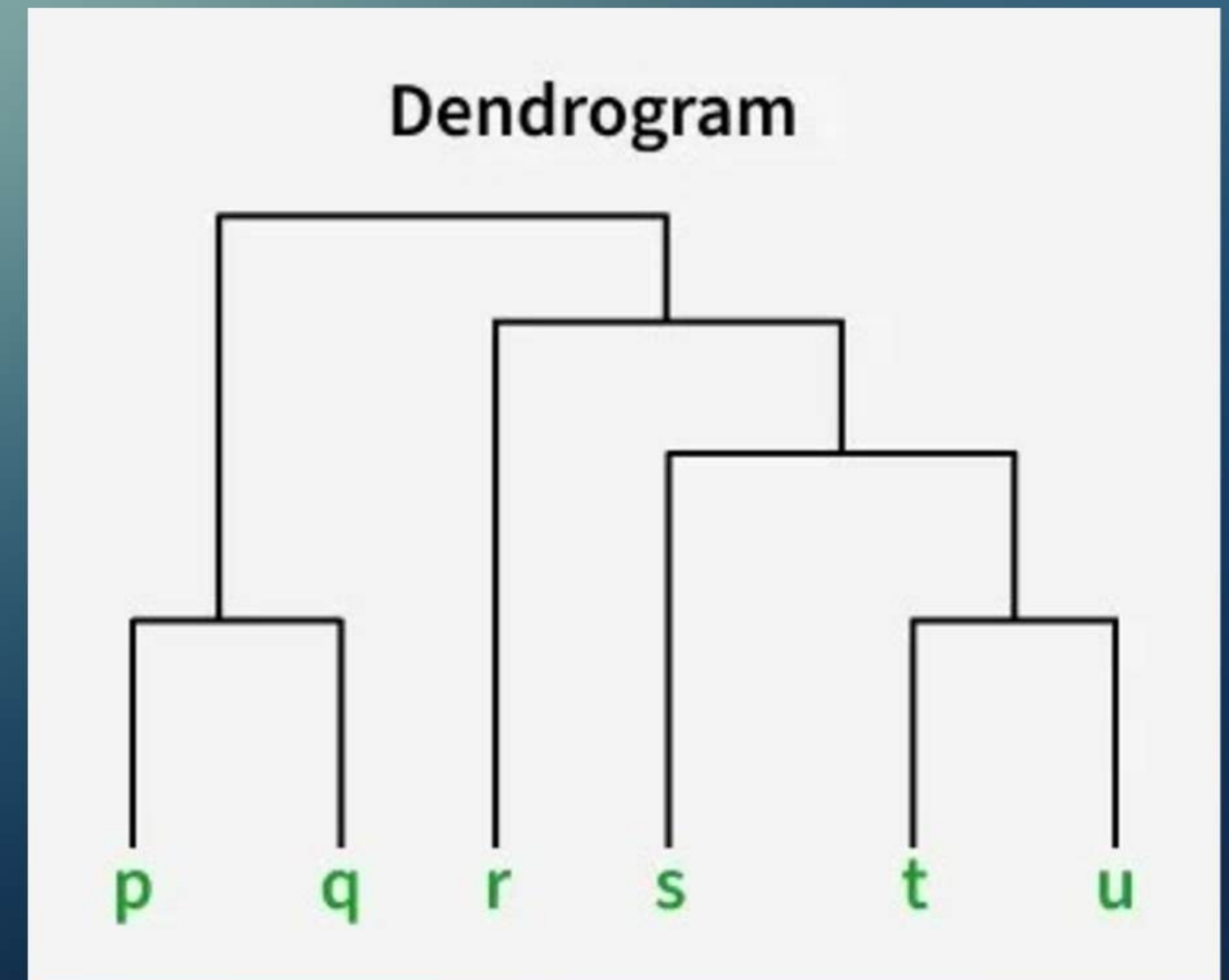
**Purpose:** Constructs clusters through successive merging based on similarity

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## Agglomerative

Bottom up approach

- Starts with each observation being own cluster
  - Decision needed on how to merge them
- Non-parametric approach
- Produces a dendrogram representing the merging process
  - Need to decide on where to “cut” tree to form clusters



Source: GeeksforGeeks (2026)



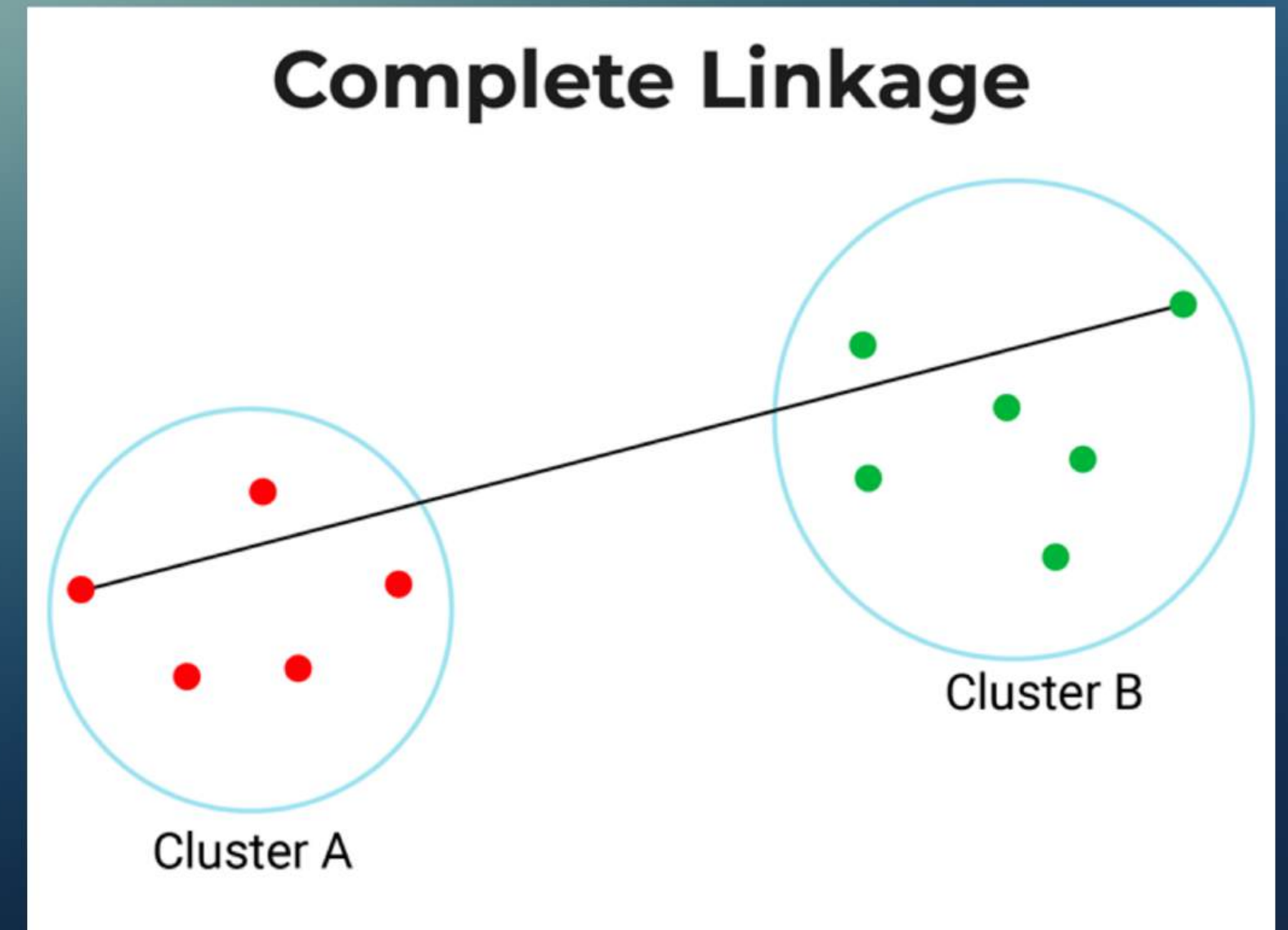
# Hierarchical Clustering

## Linkage

Decision on how to merge clusters

- Complete
  - Measures how far apart the most separated points are across the two clusters.
    - Used on proportions

$$d_{\text{complete}}(C_k, C_l) = \max_{i \in C_k, j \in C_l} d(i, j),$$





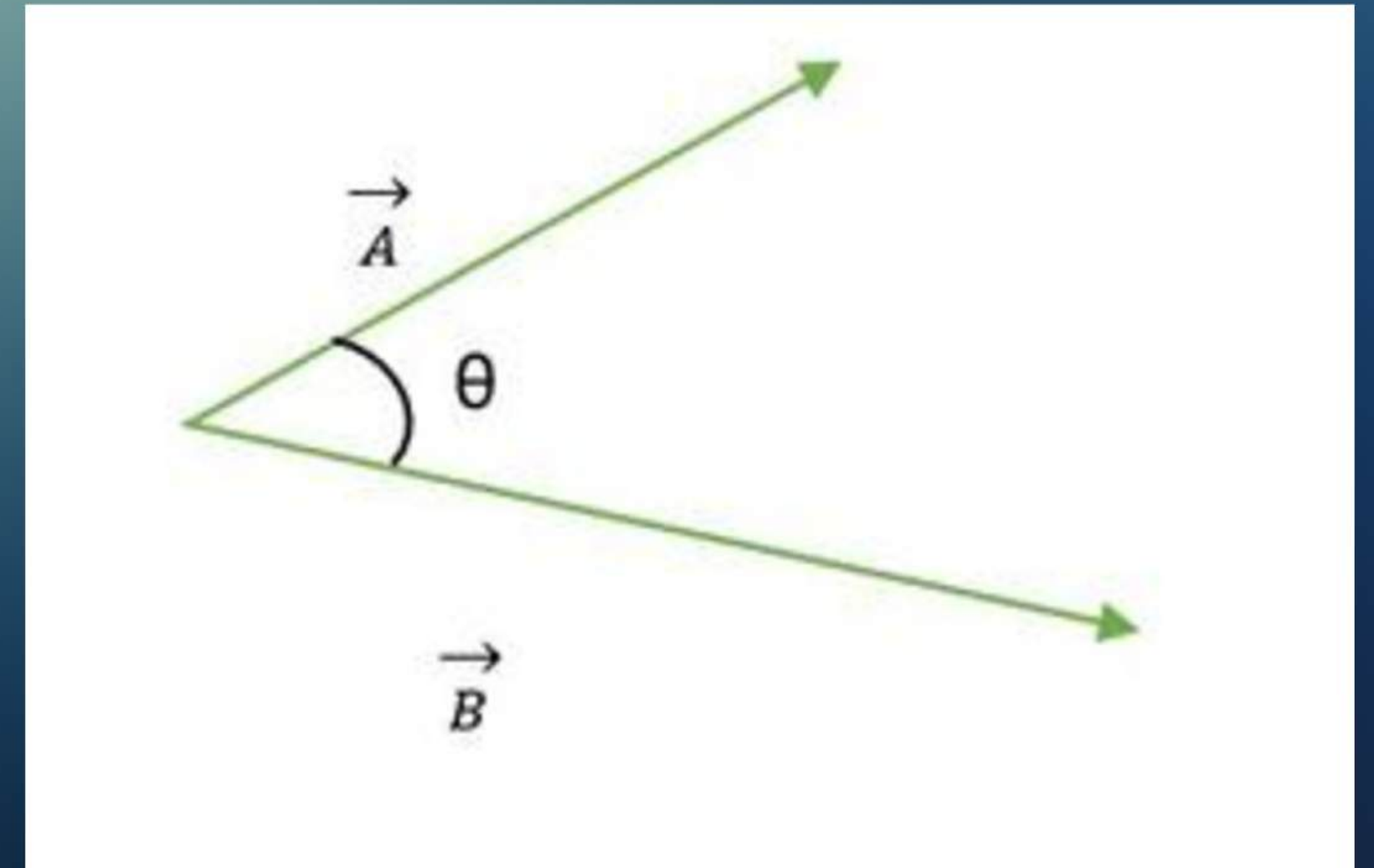
# Disimilarity Measures

## Angular Distance

- Angle between two vectors
  - Composition based

$$d_{\text{ang}}(x_i, x_j) = \arccos \left( \frac{x_i^\top x_j}{\|x_i\| \|x_j\|} \right)$$

- Measures similarity in relative proportions by comparing vector direction
  - Invariant to magnitude
- Overview of similar dates
  - Does not consider differences in product proportions



Source: GeeksforGeeks (2025)



# Silhouette Method

**Purpose:** Determine optimal number of clusters; where to “cut” structure

## Silhouette Width

- Measures how well a point fits within its cluster
  - Compares its average distance to points in its own cluster vs. the nearest other cluster

$a(i)$  = average distance within its cluster,  
 $b(i) = \min_{C_k \neq C_i}$  average distance to other clusters,

where

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \quad -1 \leq s(i) \leq 1.$$

## Silhouette Index

- Summarizes overall quality

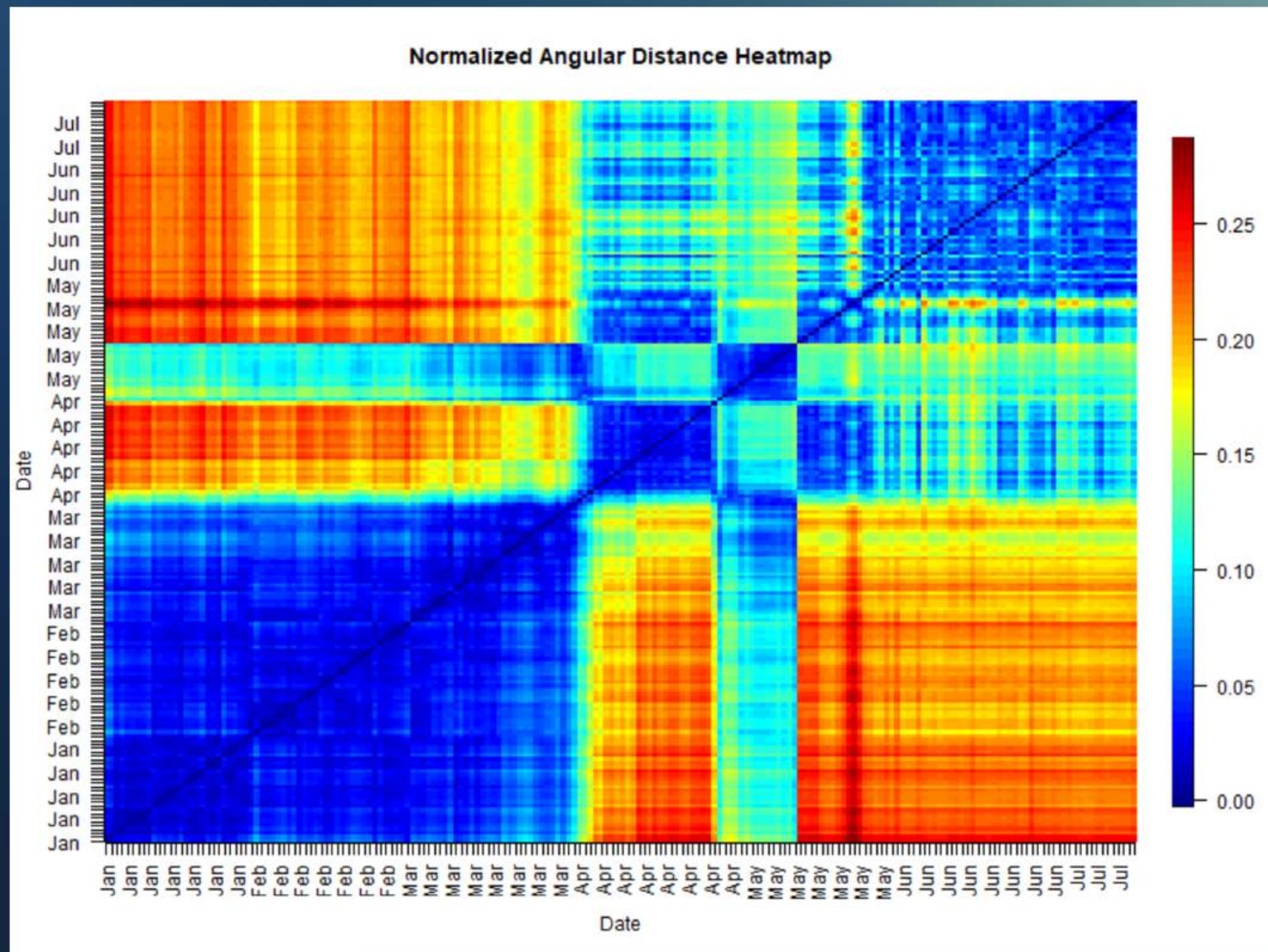
$$S = \frac{1}{n} \sum_{i=1}^n s(i)$$



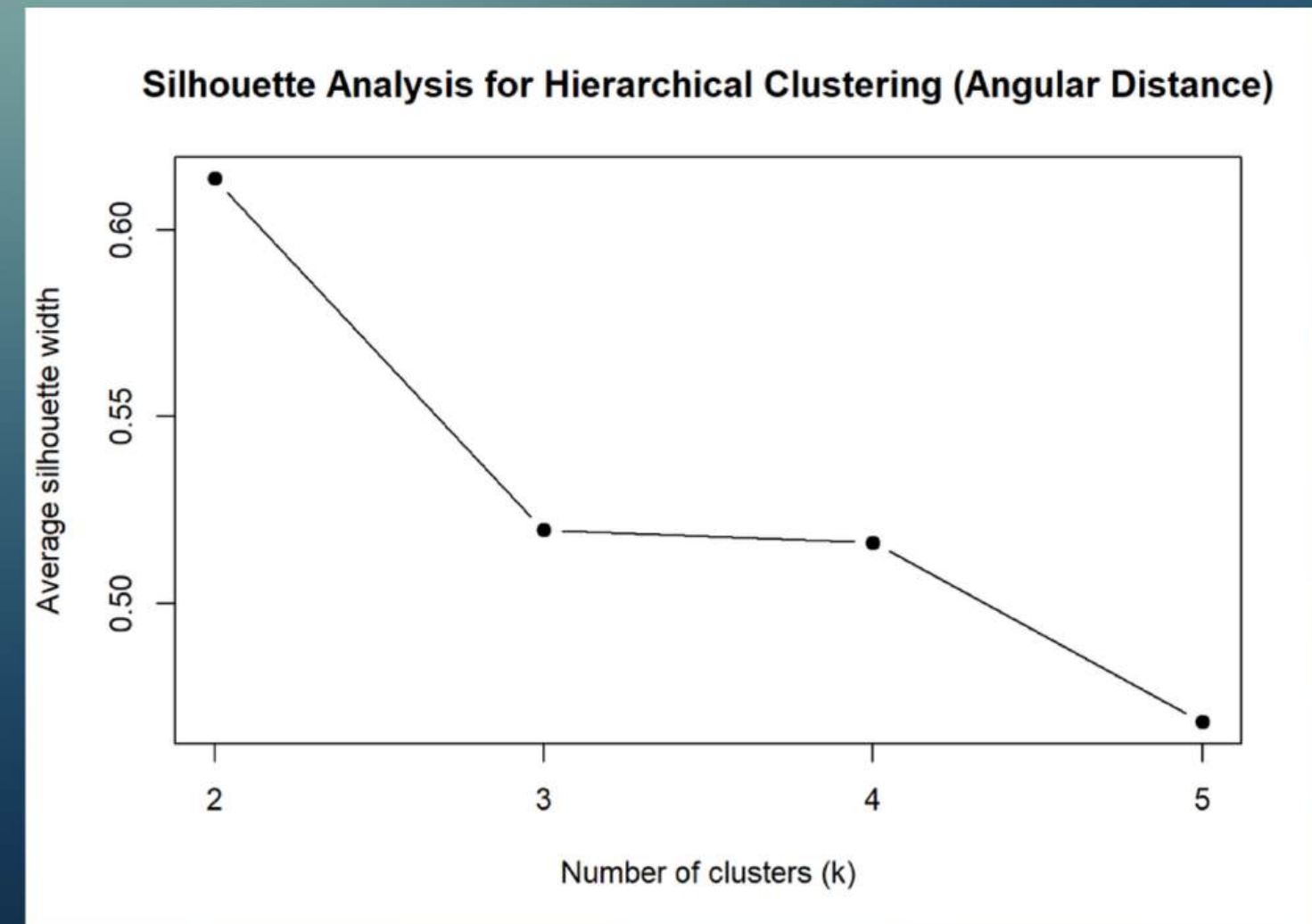
# Product Preferences

**Purpose:** Cluster dates based on vector direction

## Angular Distance Dissimilarity Matrix



## Silhouette Index by Cluster



**Note:** Complete Linkage used

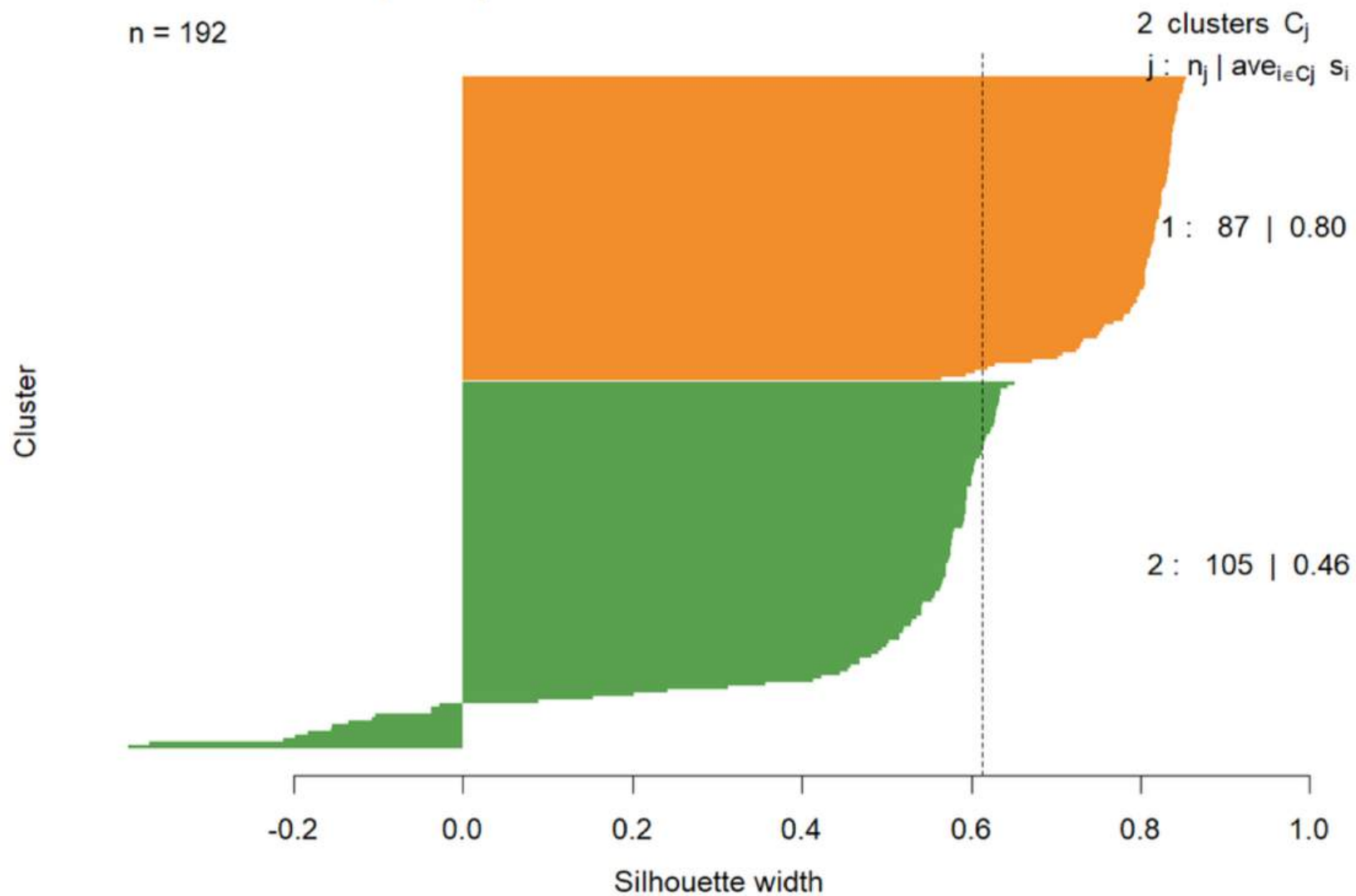


# Product Preferences

## Silhouette Scores

Silhouette Plot (k = 2)

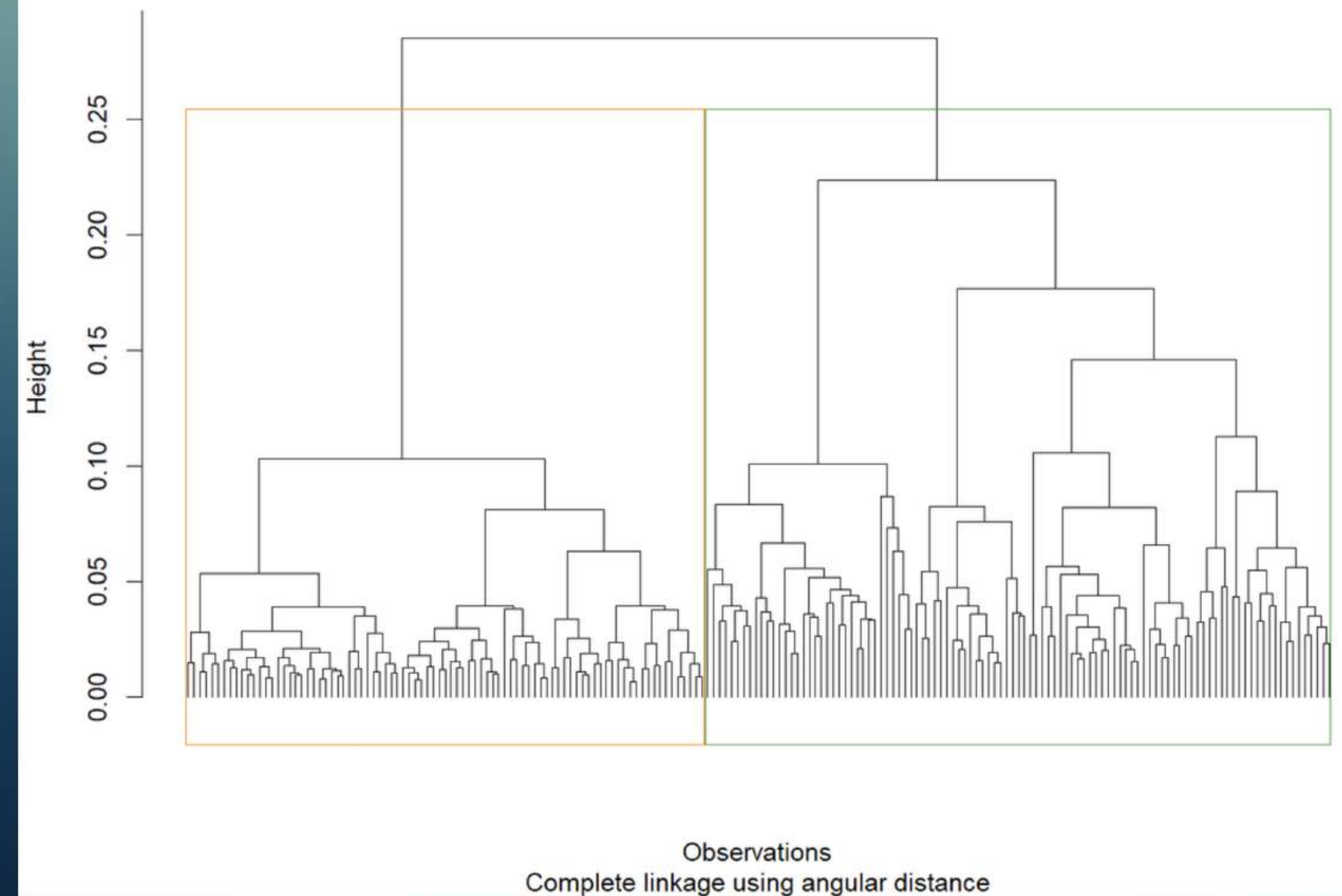
n = 192



Average silhouette width : 0.61

## Dendrogram

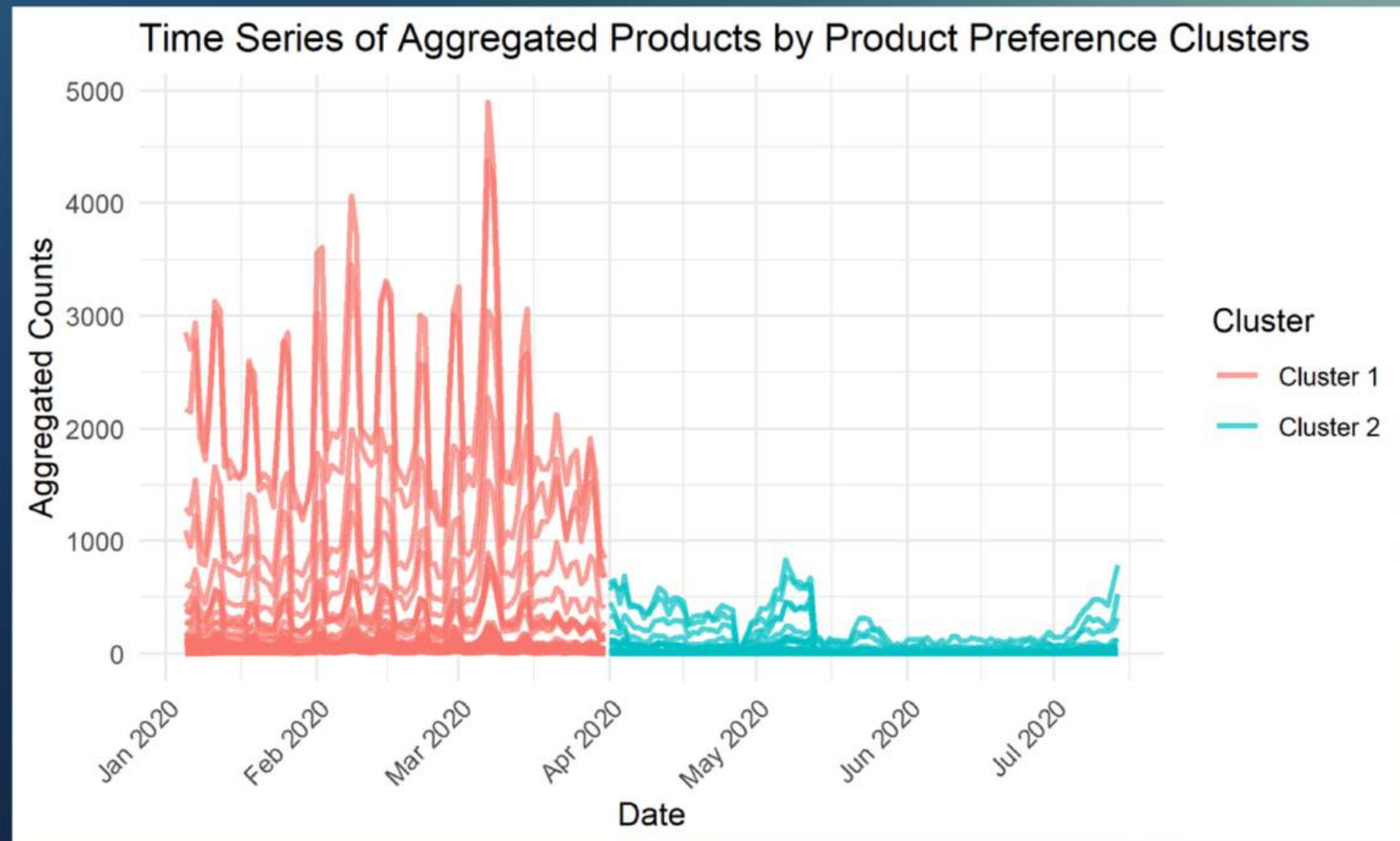
Hierarchical Clustering Dendrogram





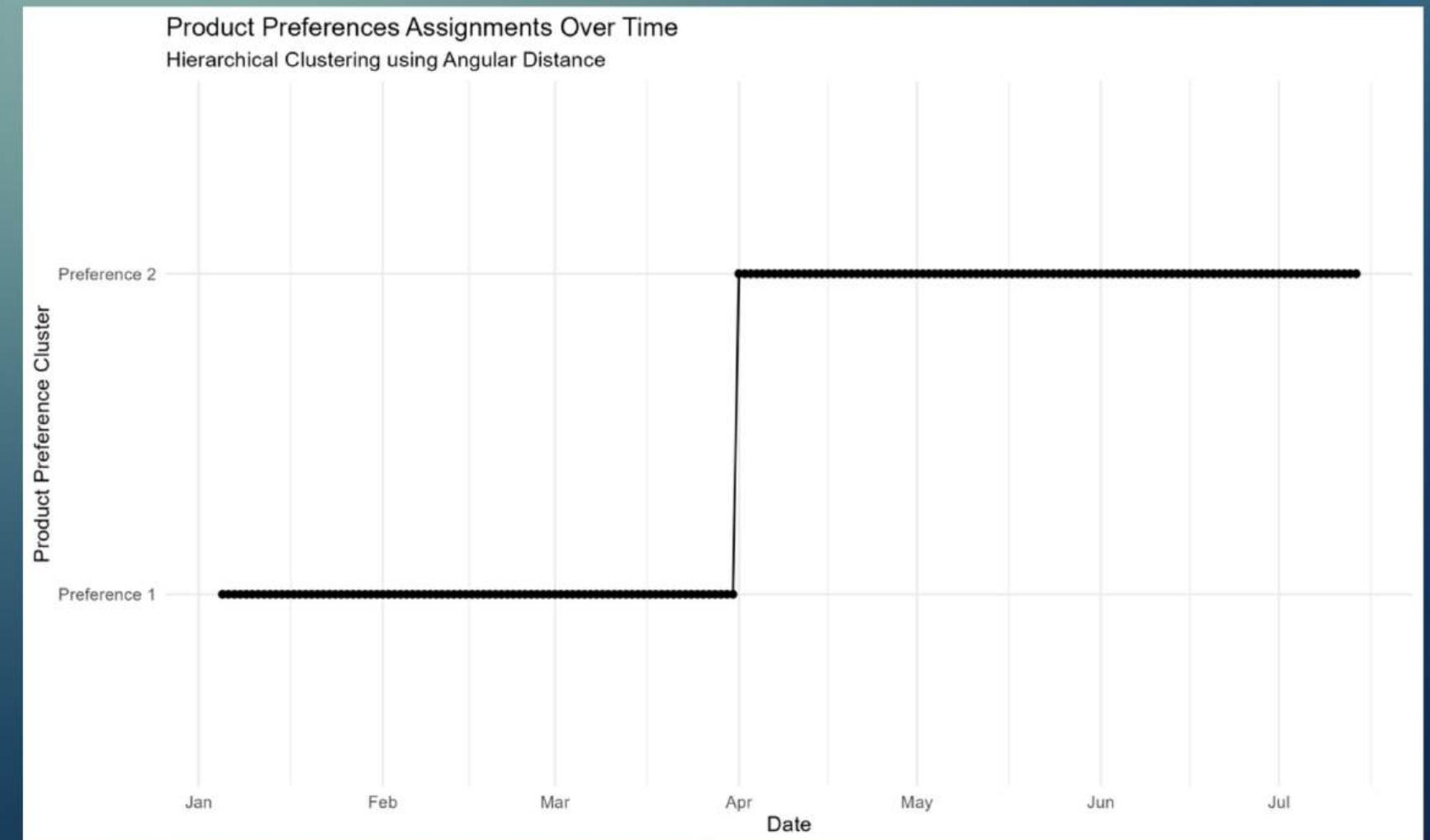
# Product Preferences

## Cluster Assignments



- Transition occurred April 1st

## Product Preferences



- Cluster 1: 87 days (45%)
- Cluster 2: 105 days (55%)



# Disimilarity Measures

## Kullback-Liebler Divergence



## Jensen-Shannon Distance

- Measures the information loss from assuming the wrong distribution
  - Not symmetric

$$D_{\text{KL}}(\theta_i || \theta_l) = \sum_{j=1}^p \theta_{i,j} \log \left( \frac{\theta_{i,j}}{\theta_{l,j}} \right)$$

- Dissimilarity between probability distributions
  - Element based
- Symmetric and bounded
  - Satisfies triangle inequality

$$d_{\text{JS}}(\theta_i, \theta_l) = \sqrt{\frac{1}{2} D_{\text{KL}}(\theta_i || m) + \frac{1}{2} D_{\text{KL}}(\theta_l || m)}$$

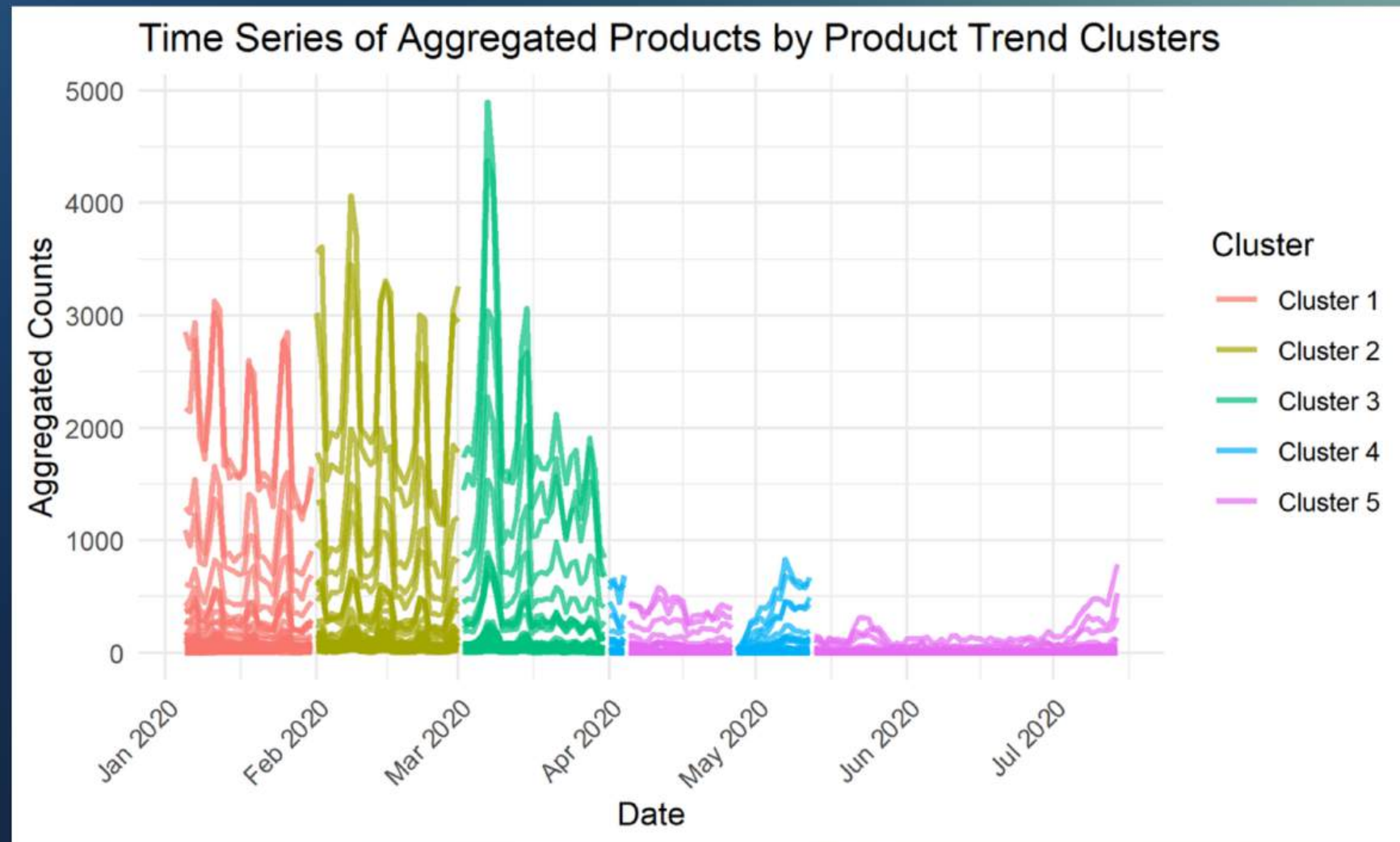
$$m = \frac{1}{2}(\theta_i + \theta_l)$$



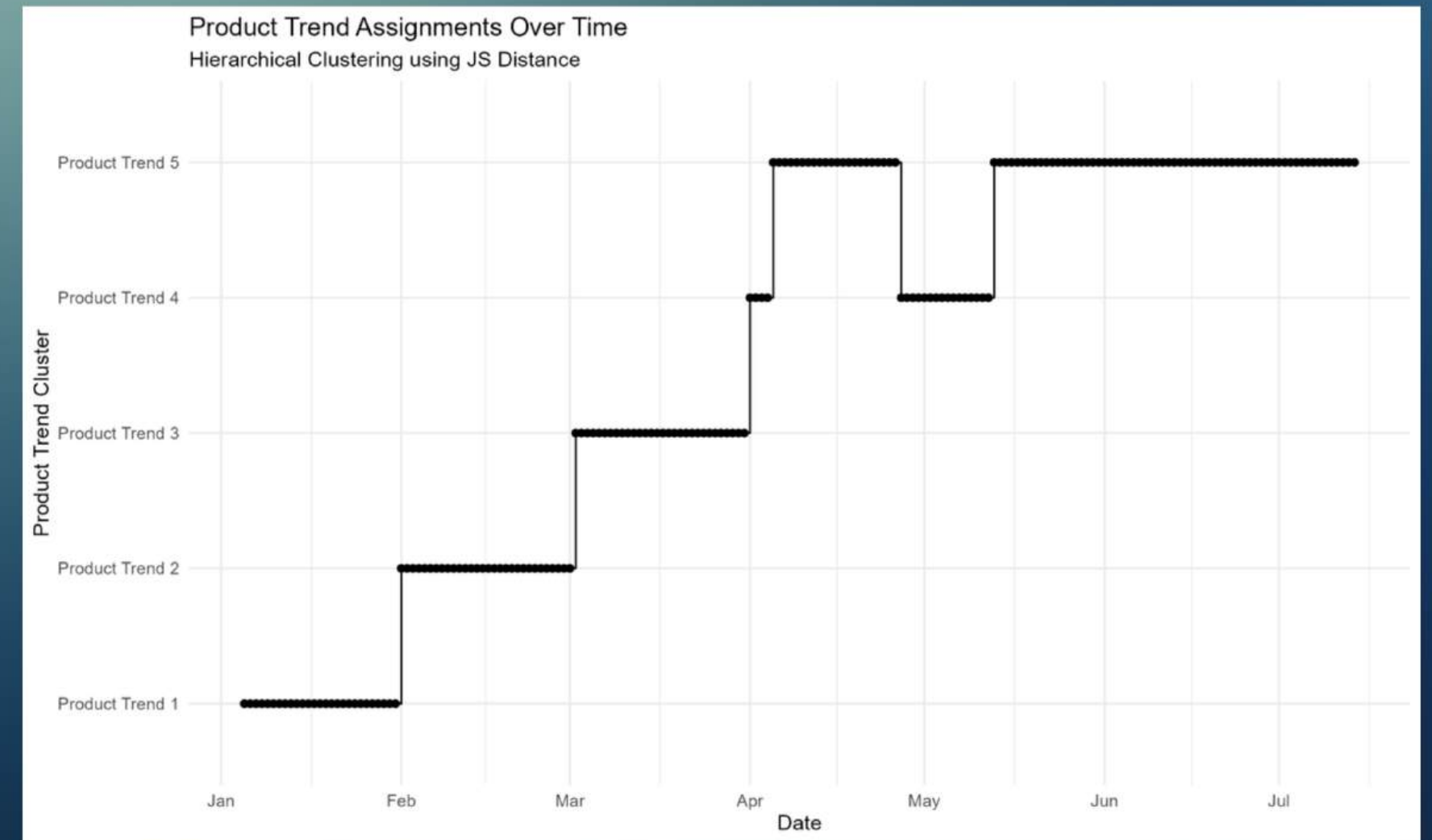
# Product Trends

**Purpose:** Cluster similar dates by similar product proportions

## Cluster Assignments



## Product Trends



**Note:** Complete Linkage used

# Product Trends

## Summary

| Product Trend | Count | Proportion (%) | Months Occurred |
|---------------|-------|----------------|-----------------|
| 1             | 27    | 14.1           | Jan             |
| 2             | 30    | 15.6           | Feb             |
| 3             | 30    | 15.6           | Mar             |
| 4             | 18    | 9.4            | Apr–May         |
| 5             | 85    | 44.3           | Apr–Jul         |

Table 3.4: Product Trend Profiles



# Dirichlet Distribution

**Purpose:** Model similar dates to understand underlying distribution

## Properties

- Multivariate case for probabilities
- Assumes independence
  - Clustering mitigates temporal dependence
- Alpha parameters describe shape and concentration

$$\hat{\theta}_i \mid z_i = k \sim \text{Dirichlet}(\alpha^k),$$

where  $\alpha^k = (\alpha_1^k, \dots, \alpha_p^k)$  with  $\alpha_j^k > 0$

$$\alpha_j^k = \mu_{k,j} \kappa_k,$$

$$\kappa_k = \sum_{j=1}^p \alpha_j^k$$

$$\mu_k = \frac{\alpha^k}{\sum_{j=1}^p \alpha_j^k}$$

$$\mathbb{E}[\hat{\theta}_i \mid z_i = k] = \mu_k.$$



# Modeling Product Trends

**Purpose:** Approximate the underlying distribution from which dates are drawn

## Parameters

| Trend   | $\alpha < 1$      | $1 \leq \alpha < 2$ | $2 \leq \alpha < 10$ | $\alpha \geq 10$  | $\kappa_k$ |
|---------|-------------------|---------------------|----------------------|-------------------|------------|
| Trend 1 | 0 (0.0%)          | 0 (0.0%)            | 24 (49.0%)           | <b>25 (51.0%)</b> | 5050.60    |
| Trend 2 | 0 (0.0%)          | 0 (0.0%)            | 25 (51.0%)           | <b>24 (49.0%)</b> | 3958.60    |
| Trend 3 | 0 (0.0%)          | 0 (0.0%)            | 27 (55.1%)           | <b>22 (44.9%)</b> | 3894.50    |
| Trend 4 | 0 (0.0%)          | 0 (0.0%)            | <b>39 (79.6%)</b>    | 10 (20.4%)        | 1977.79    |
| Trend 5 | <b>21 (42.9%)</b> | 10 (20.4%)          | 16 (32.7%)           | 2 (4.1%)          | 286.20     |

Table 3.5: Distribution of Dirichlet concentration parameters across thresholds for all trends.

## Differences Between Trends

|         | Trend 1 | Trend 2 | Trend 3 | Trend 4 | Trend 5 |
|---------|---------|---------|---------|---------|---------|
| Trend 1 | 0.0000  | 0.0629  | 0.0840  | 0.1682  | 0.3827  |
| Trend 2 | 0.0629  | 0.0000  | 0.0845  | 0.1589  | 0.3690  |
| Trend 3 | 0.0840  | 0.0845  | 0.0000  | 0.1167  | 0.3379  |
| Trend 4 | 0.1682  | 0.1589  | 0.1167  | 0.0000  | 0.2556  |
| Trend 5 | 0.3827  | 0.3690  | 0.3379  | 0.2556  | 0.0000  |

Table 3.6: Pairwise Jensen–Shannon distances between trend mean distributions

- Trends are highly concentrated around their mean composition

- Separation between clusters
  - Trends 1-3 similar, Trend 5 least similar



# Sequence Analysis

**Purpose:** Understand how cluster assignments change over time

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## Transitions

- Probability of going from one assignment to another
  - Predict next state based on current

$$P_{ij} \approx \frac{\text{Count}\{t : z_t = i, z_{t+1} = j\}}{\text{Count}\{t : z_t = i\}}$$

## N-grams

- Identifying longer term patterns
  - Reoccurring weeks

$$P(z_t \mid z_{t-n+1:t-1}) \approx \frac{\text{Count}(z_{t-n+1:t})}{\text{Count}(z_{t-n+1:t-1})}$$

# Product Behavior

**Purpose:** Understand product trends over time

## Comparing Trends

| Trend   | 0% | 1–5% | 5%+ |
|---------|----|------|-----|
| Trend 1 | 0  | 44   | 5   |
| Trend 2 | 0  | 44   | 5   |
| Trend 3 | 0  | 44   | 5   |
| Trend 4 | 0  | 44   | 5   |
| Trend 5 | 9  | 36   | 4   |

Table 3.7: Distribution of Product Proportions Within Each Trend

## Product Behavior

| Product | T1    | T2    | T3    | T4    | T5    | Behavior   |
|---------|-------|-------|-------|-------|-------|------------|
| 286     | 20.6% | 18.7% | 17.7% | 10.8% | 0.4%  | Decline    |
| Others  | 19.7% | 20.9% | 20.5% | 20.2% | 27.3% | Increase   |
| 1       | 10.9% | 11.0% | 13.9% | 18.3% | 21.1% | Increase   |
| 3       | 8.6%  | 7.9%  | 9.0%  | 11.8% | 14.1% | Increase   |
| 5       | 5.5%  | 6.1%  | 5.8%  | 6.3%  | 7.0%  | Increase   |
| 308     | 1.7%  | 1.5%  | 0.6%  | 0.6%  | 0.0%  | Decline    |
| 305     | 1.2%  | 1.2%  | 1.0%  | 0.9%  | 0.0%  | Decline    |
| 62      | 0.3%  | 0.3%  | 0.3%  | 0.3%  | 0.4%  | Consistent |

Table 3.8: Product means across trends



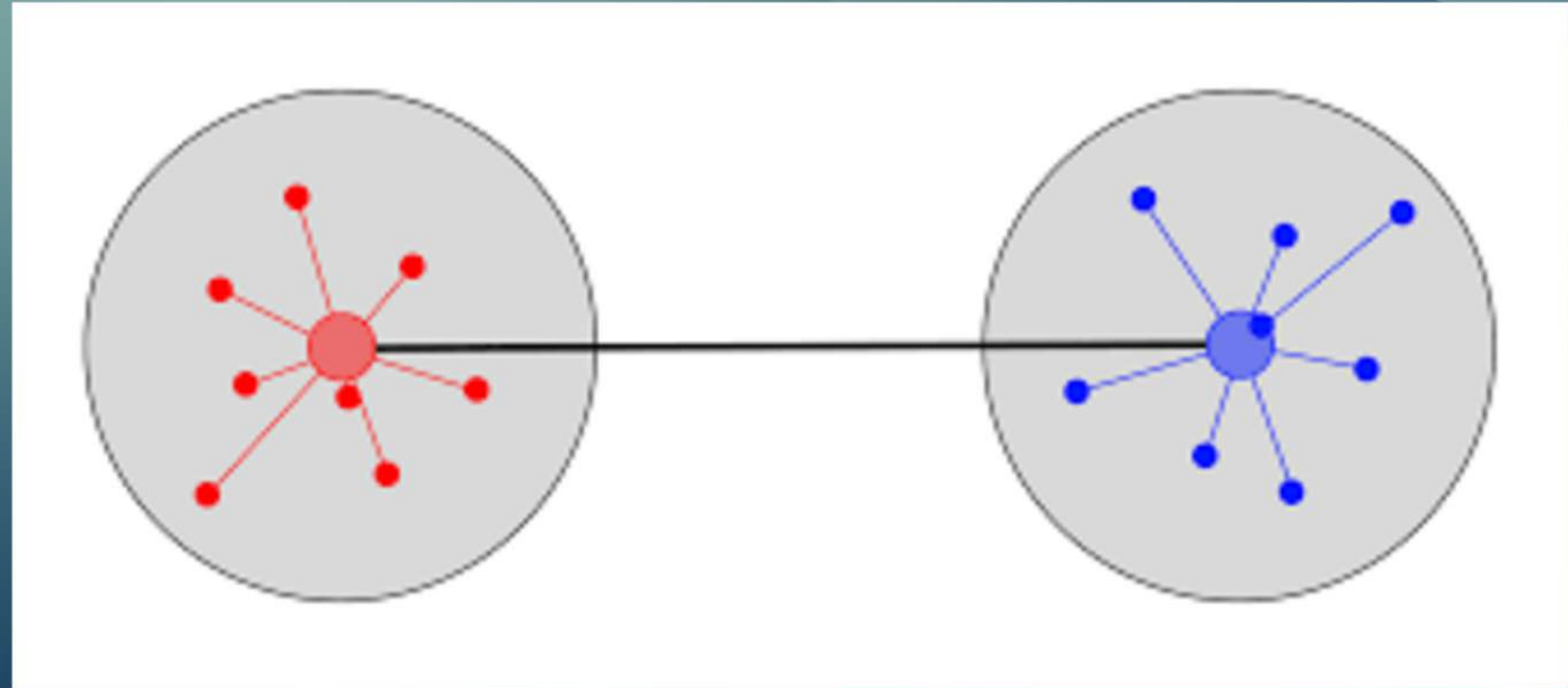
# Hierarchical Clustering

## Linkage

- Ward's Method
  - Defines cluster distance as the increase in total within-cluster variance
    - Used for aggregated counts

$$E = \sum_{k=1}^K E_k,$$

$$E_k = \sum_{i \in C_k} \sum_{j=1}^p (x_{i,j} - \bar{x}_{k,j})^2.$$



**Source: GraphPad Prism (2019)**

# Disimilarity Measures

## Euclidean Distance

- Measures how far points are from one another
- Use log-transform
  - Reduces skewness
  - Less sensitive to outliers

$$d_{\text{Euc}}(y_i, y_j) = \sqrt{\sum_{l=1}^p (y_{i,l} - y_{j,l})^2}$$

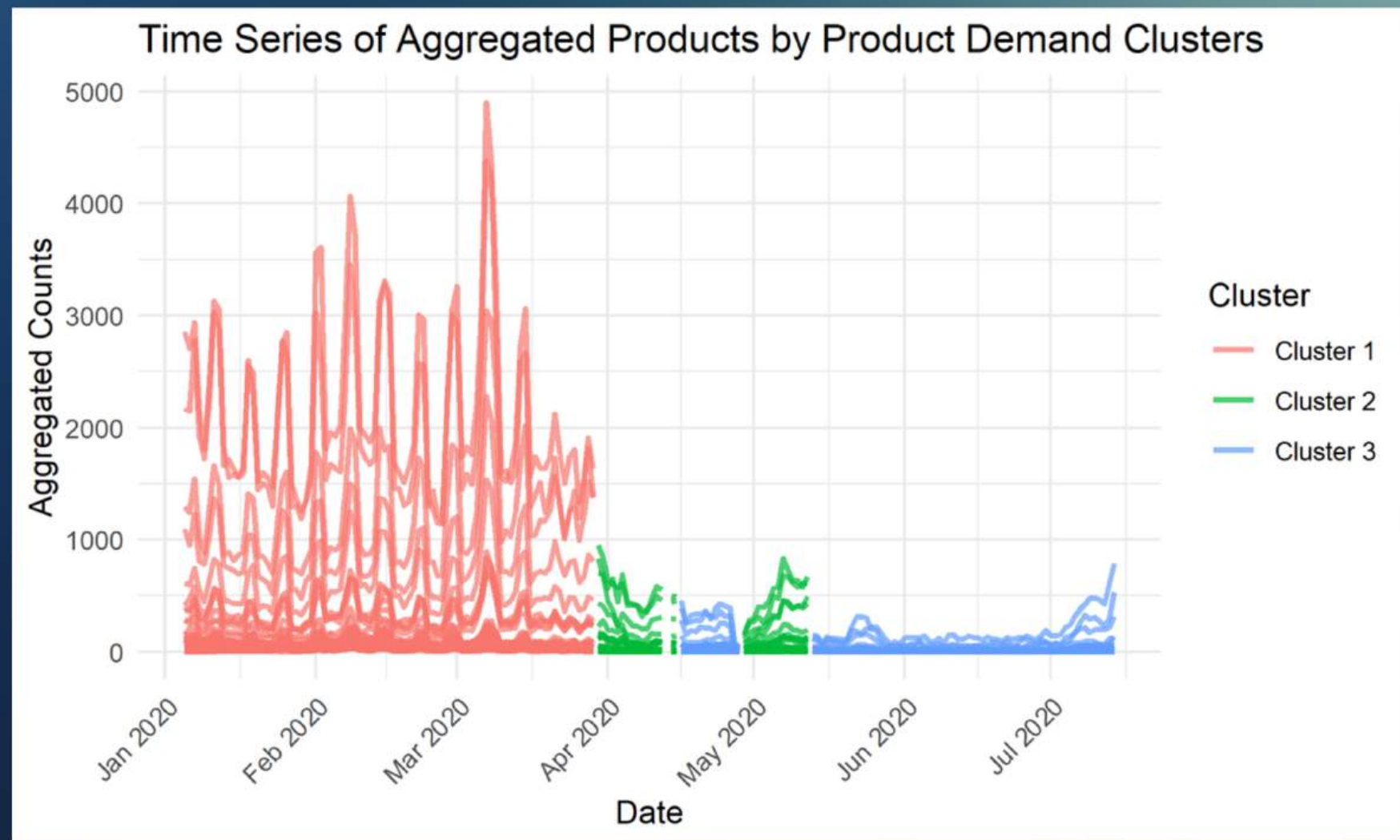
$$y_i = \log(x_i + 1)$$



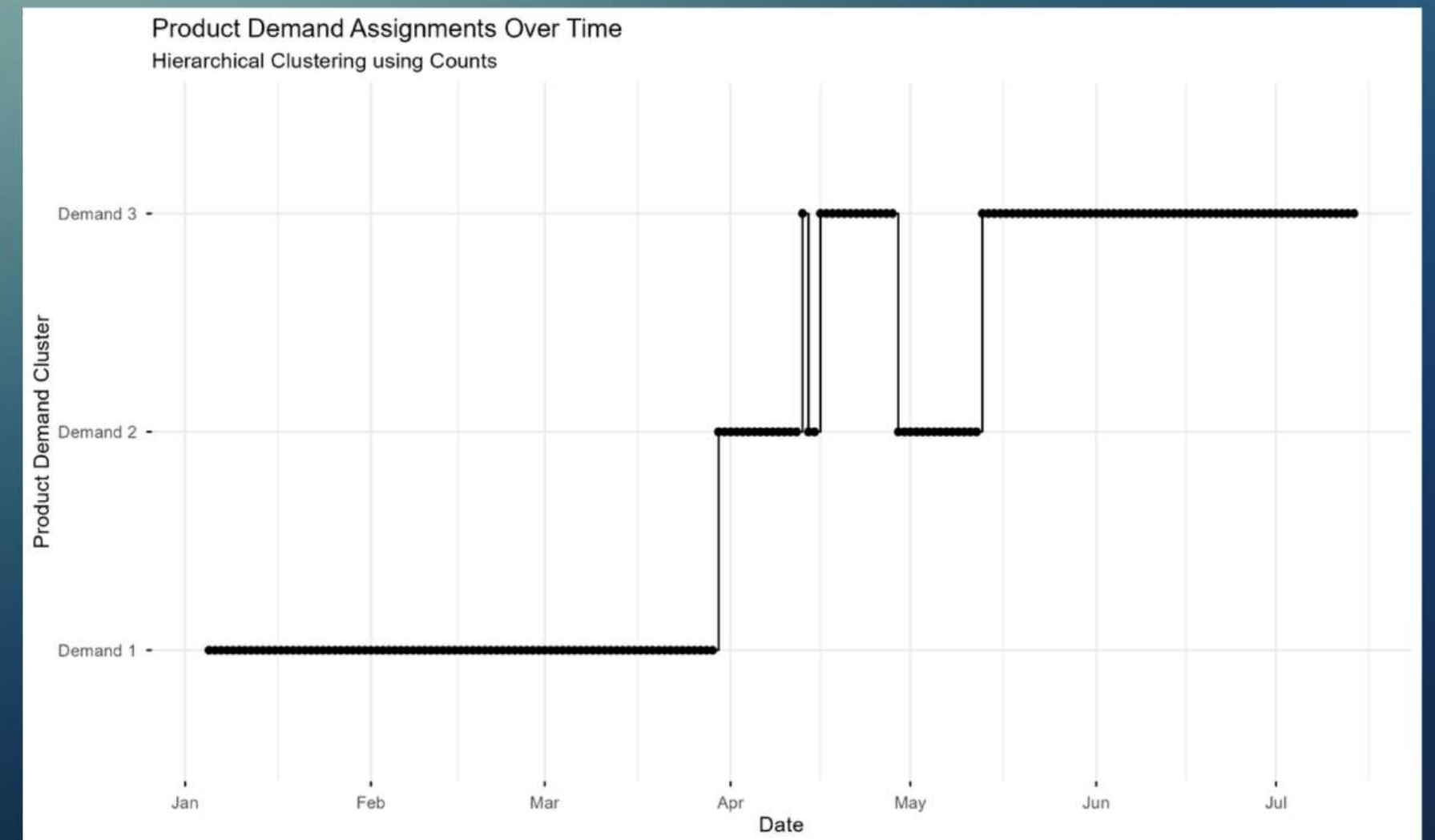
# Product Demand

**Purpose:** Cluster similar dates by aggregated product counts

## Cluster Assignments



## Product Demand



- Transition occurred March 30<sup>th</sup>

# Product Demand

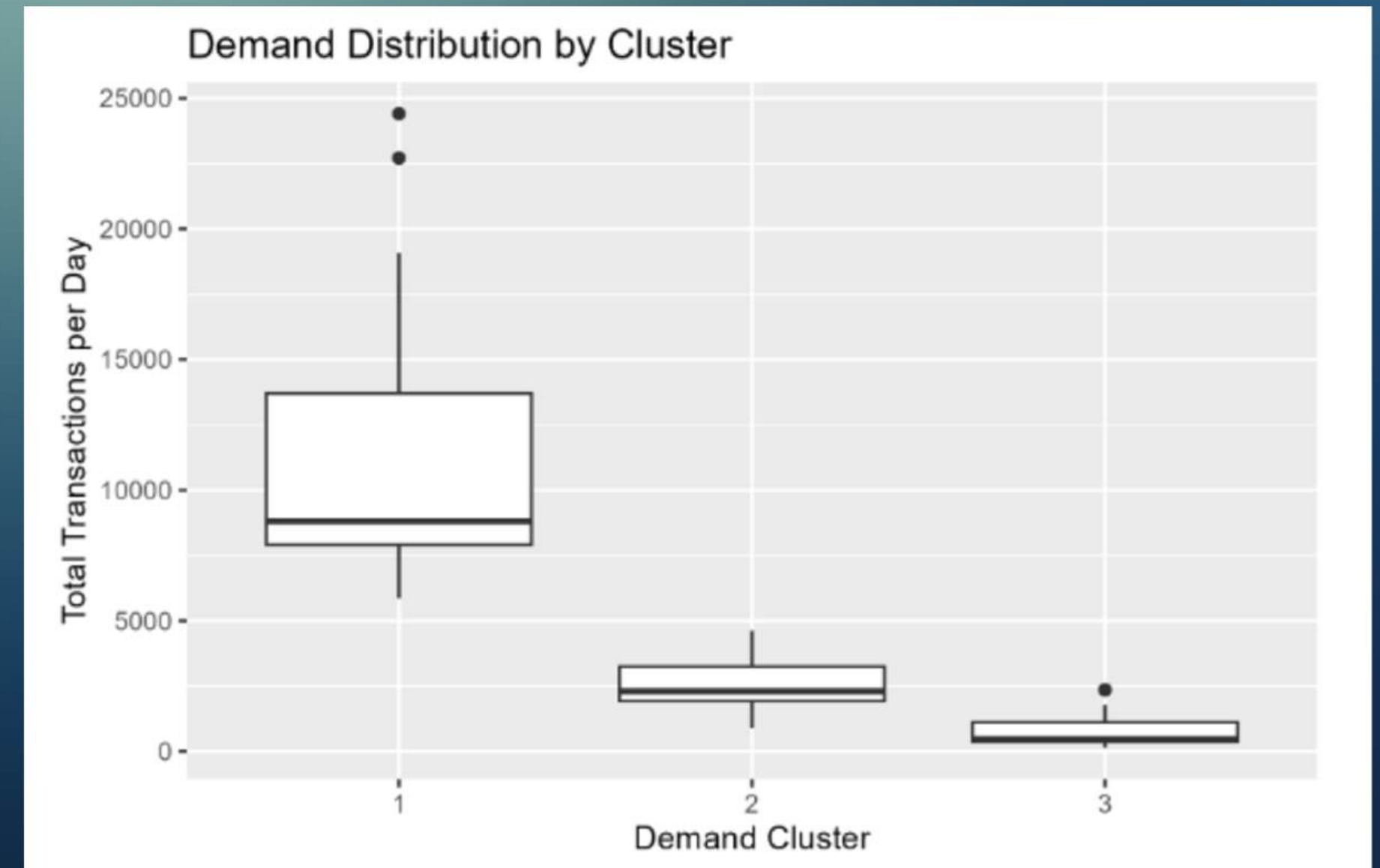
## Summary

Table 3.9: Product Demand Summary

| Product Demand | Count | Proportion (%) | Months Occurred |
|----------------|-------|----------------|-----------------|
| 1              | 85    | 44.3           | Jan–Mar         |
| 2              | 30    | 15.6           | Apr–May         |
| 3              | 77    | 40.1           | Apr–Jul         |

- Change in demand occurred before change in preference
  - CISA updated essential worker guidelines Mar 28th

## Distribution by Cluster







# Results

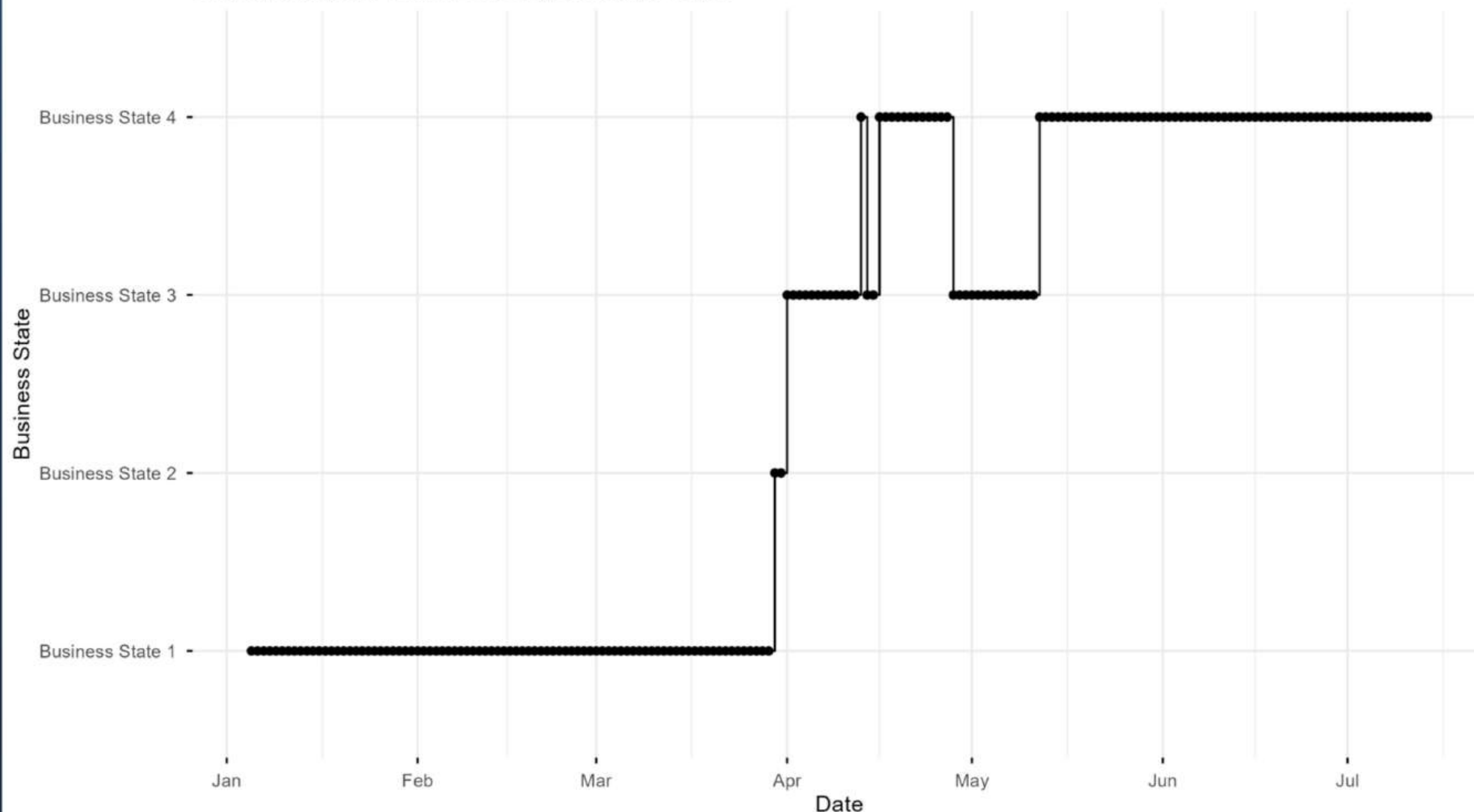


# Business States

$$B_b = \{x_i \mid z_i^{(preferences)} = m, z_i^{(demand)} = d\}$$

## Business States

Business State Assignments Over Time  
Constructed from Product States and Demand States



## Summary

Table 4.1: Business State Summary

| Business State | Count | Proportion (%) | Months Occurred |
|----------------|-------|----------------|-----------------|
| 1              | 85    | 44.3           | Jan-Mar         |
| 2              | 2     | 1.0            | Mar             |
| 3              | 28    | 14.6           | Apr-May         |
| 4              | 77    | 40.1           | Apr-Jul         |

- Decrease in demand, then change in product preferences
  - 9 out of stock products



# Business States

## Transitions Summary

| Business State Transition | States Involved   | Behavior                      |
|---------------------------|-------------------|-------------------------------|
| Transition 1              | State 1 → State 2 | Decrease in demand            |
| Transition 2              | State 2 → State 3 | Change in product preferences |
| Transition 3              | State 3 → State 4 | Decrease in demand            |
| Transition 4              | State 4 → State 3 | Increase in demand            |

Table 4.2: Summary of business state transitions

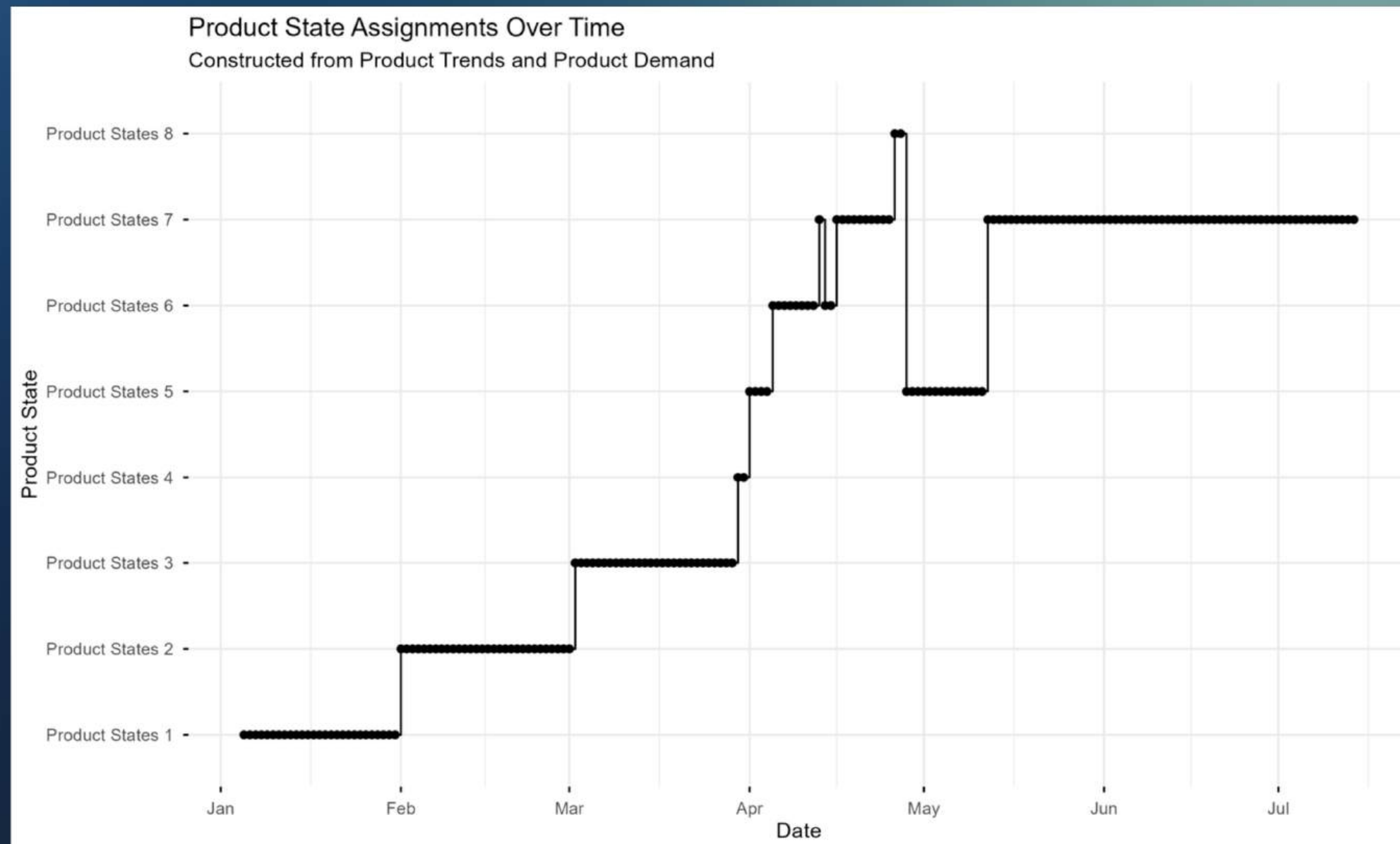
## Key Finding

- Suspected inventory disruption
  - Labor shortage
  - Canceled Shipments

# Product States

$$P_r = \{x_i \mid z_i^{(trends)} = g, z_i^{(demand)} = h\}$$

## Product States



## Summary

| Table 4.3: Product State Summary |       |                |                 |
|----------------------------------|-------|----------------|-----------------|
| Product State                    | Count | Proportion (%) | Months Occurred |
| 1                                | 27    | 14.1           | Jan             |
| 2                                | 30    | 15.6           | Jan-Feb         |
| 3                                | 28    | 14.6           | Feb-Mar         |
| 4                                | 2     | 1.0            | Mar             |
| 5                                | 18    | 9.4            | Apr-May         |
| 6                                | 10    | 5.2            | Apr             |
| 7                                | 75    | 39.1           | May-Jul         |
| 8                                | 2     | 1.0            | Jun             |

- Change in customer purchasing behavior
  - Jan-Mar



# Product States

## Transitions Summary

| Transition    | States Involved   | Cause of Change in Behavior |
|---------------|-------------------|-----------------------------|
| Transition 1  | State 1 → State 2 | Product Trends              |
| Transition 2  | State 2 → State 3 | Product Trends              |
| Transition 3  | State 3 → State 4 | Overall Demand Decrease     |
| Transition 4  | State 4 → State 5 | Product Trends              |
| Transition 5  | State 5 → State 6 | Product Trends              |
| Transition 6  | State 6 → State 7 | Overall Demand Decrease     |
| Transition 7  | State 7 → State 6 | Overall Demand Increase     |
| Transition 8  | State 7 → State 8 | Product Trends              |
| Transition 9  | State 8 → State 5 | Overall Demand Increase     |
| Transition 10 | State 5 → State 7 | Overall Demand Decrease     |

Table 4.4: Summary of Product States Transitions

## Key Findings

- Change in product trends leading up to suspected disruption
  - Product 286 declined
- Change in available products caused change in purchasing patterns

# Conclusion



# Summary of Findings

- **Product preferences:** 2 clusters found
- **Product trends:** 5 clusters found
- **Product behavior:** Decrease in product 286
- **Product demand:** 3 clusters found, supply disruptions influenced customer purchasing patterns
- **Business States:** 4 states found; transitions driven by inventory
- **Product States:** 5 states found; changes in trends leading up to suspected disruption, inventory shortage suspected to cause further change in trends

# Unexpected Findings

- **Consistent cluster and state assignments**
  - No transition back to earlier states
  - No use of n-grams
  - No further use of temporal features
    - Day of week
- **Overlap between clusters**
  - Product demand showed signs of degeneracy
    - Conservative approach to determining clusters



# Insights

- **Business Performance**

- Product preferences and product demand consistent until suspected disruption
  - Once disruption occurred, labor and inventory disrupted operations

- **Recommendations**

- Operations
  - Use business states found to staff accordingly and potentially where
- Inventory Strategy
  - Knowing product states can determine how much inventory to stock and adjust as change in product behavior occurs.
- Marketing Initiatives
  - Product trends can be used as a baseline to measure new initiatives

# Future Work

- **Dynamic Tree Cut**
  - Flexible approach to determining clusters
- **Business Context**
  - Apply approach to data with product descriptions





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# Thank You



# References

- Kechinov, M. (2020). eCommerce purchase history from electronics store. Kaggle. <https://www.kaggle.com/datasets/mkechinov/ecommerce-purchase-history-from-electronics-store>
- Corander, J., Remes, U., & Koski, T. (2021). On the Jensen–Shannon divergence and the variation distance for categorical probability distributions. *Kybernetika*, 57(6), 879–907. <https://doi.org/10.14736/kyb-2021-6-0879>
- Everitt, B. S., Landau, S., Leese, M., & Stahl, D. (2011). *Cluster analysis* (5th ed.). Wiley.
- Kumar, S. D., & Santhosh, A. M. (2024). The impact of inventory management on business performance: A comprehensive study. *International Journal of Research in Finance and Management*, 7(2), 592–595.
- Lin, J. (2016). On the Dirichlet distribution (Master's thesis, Queen's University). <https://mast.queensu.ca/~communications/Papers/msc-jiayu-lin.pdf>
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
- Salamin, H., & Vinciarelli, A. (2011). Introduction to sequence analysis for human behavior understanding. In A. Vinciarelli, M. Pantic, & H. Bourlard (Eds.), *Social signal processing* (pp. 165–185). Cambridge University Press.
- Van Ruitenbeek, R. E., Koole, G. M., & Bhulai, S. (2023). A hierarchical agglomerative clustering for product sales forecasting. *Decision Analytics Journal*, 8, 100318.